

# Day 1 — Data Analyst Placement Practice

## Foundation & Diagnostic | Full Question Paper

Based on the Day 1 syllabus: SQL (SELECT/WHERE/ORDER BY), Python (variables/loops/conditionals), Number System, Table-based Data Interpretation, Error Spotting, Coding-Decoding, Psychometric speed round, and a short Reading Comprehension passage.

| Section                     | Questions               | Suggested Time   |
|-----------------------------|-------------------------|------------------|
| SQL                         | 3                       | 10 min           |
| Python                      | 1                       | 5 min            |
| Numeric — Number System     | 3                       | 7 min            |
| Data Interpretation — Table | 3                       | 8 min            |
| Verbal — Error Spotting     | 4                       | 6 min            |
| Reasoning — Coding-Decoding | 5                       | 8 min            |
| Psychometric                | 10                      | 2 min HARD LIMIT |
| Reading Comprehension       | 1 passage + 3 questions | 7 min            |

Instructions: Attempt first. Do not look for solutions while answering. Record your selected option for every question.

### Section A — SQL (3 Questions)

**Q1.** Given the table **employees(emp\_id, name, department, salary)**, which query returns the names and salaries of employees in the IT department whose salary is greater than 60000, sorted from highest to lowest salary?

- A. `SELECT name, salary FROM employees WHERE department = 'IT' AND salary > 60000 ORDER BY salary DESC;`
- B. `SELECT name, salary FROM employees WHERE department = 'IT' OR salary > 60000 ORDER BY salary ASC;`
- C. `SELECT name, salary FROM employees WHERE department = 'IT' AND salary >= 60000 ORDER BY salary DESC;`
- D. `SELECT name, salary FROM employees WHERE salary > 60000 ORDER BY salary DESC;`

**Q2.** What is the result of the following query on **products(id, name, price)**?

`SELECT name, price FROM products WHERE price BETWEEN 1000 AND 2000 ORDER BY price ASC;`

- A. Products priced below 1000, sorted ascending.
- B. Products priced from 1000 through 2000, sorted from lowest to highest.
- C. Products priced above 2000, sorted descending.
- D. All products, sorted by name.

**Q3.** Which clause is used to filter rows *before* the result is sorted, and which clause controls the sorting order?

- A. WHERE filters rows; ORDER BY sorts the result.

- B. ORDER BY filters rows; WHERE sorts the result.
- C. SELECT filters rows; WHERE sorts the result.
- D. GROUP BY filters rows; ORDER BY selects columns.

### Section B — Python (1 Question)

**Q4.** What is printed by this Python code?

```
x = 5
for i in range(3):
    x = x + i
print(x)
```

- A. 5
- B. 6
- C. 8
- D. 11

### Section C — Numeric Aptitude: Number System (3 Questions)

**Q5.** What is the remainder when 257 is divided by 13?

- A. 8
- B. 9
- C. 10
- D. 11

**Q6.** Which of the following numbers is divisible by both 3 and 5?

- A. 245
- B. 315
- C. 428
- D. 512

**Q7.** If a number leaves remainder 4 when divided by 7, what remainder will 3 times that number leave when divided by 7?

- A. 1
- B. 3
- C. 5
- D. 6

### Section D — Data Interpretation: Table (3 Questions)

Study the table and answer Q8–Q10.

| Department | Jan | Feb | Mar |
|------------|-----|-----|-----|
| IT         | 120 | 150 | 180 |
| HR         | 80  | 100 | 90  |

|         |     |     |     |
|---------|-----|-----|-----|
| Sales   | 200 | 220 | 250 |
| Finance | 100 | 120 | 140 |

**Q8.** What is the total value for Sales across the three months?

- A. 620
- B. 650
- C. 670
- D. 690

**Q9.** Which department had the highest percentage increase from January to March?

- A. IT
- B. HR
- C. Sales
- D. Finance

**Q10.** What was the total value across all departments in February?

- A. 570
- B. 590
- C. 610
- D. 630

### **Section E — Verbal: Error Spotting (4 Questions)**

**Q11.** Identify the part containing an error: (A) The manager / (B) along with his team / (C) were present / (D) at the meeting.

- A. A
- B. B
- C. C
- D. D

**Q12.** Identify the part containing an error: (A) She / (B) do not / (C) understand / (D) the requirement.

- A. A
- B. B
- C. C
- D. D

**Q13.** Identify the part containing an error: (A) Each of the candidates / (B) have submitted / (C) the application / (D) before the deadline.

- A. A
- B. B
- C. C

D. D

**Q14.** Identify the part containing an error: (A) The data / (B) were analyzed / (C) carefully by the analyst / (D) yesterday.

A. A

B. B

C. C

D. D

### **Section F — Reasoning: Coding-Decoding (5 Questions)**

**Q15.** If CAT is coded as DBU, how is DOG coded?

A. EPH

B. EOG

C. DPH

D. FPH

**Q16.** If TABLE is coded as UBCMF, how is CHAIR coded?

A. DIBJS

B. DIBIR

C. DHBJS

D. EIBJS

**Q17.** In a code, DATA is written as ATAD. How is PYTHON written using the same rule?

A. NOHTYP

B. NOHTY P

C. PHTYON

D. PYT HNO

**Q18.** If ROAD is coded as URDG by shifting every letter 3 positions forward, how is DATA coded?

A. GDWD

B. GDVC

C. FCWD

D. GDWE

**Q19.** In a certain code, 123 means 'data is useful' and 345 means 'useful for analysis'. Which digit represents 'useful'?

A. 1

B. 2

C. 3

D. 4

## Section G — Psychometric Speed Round (10 Questions / 2 Minutes)

**Hard rule:** Set a 2-minute timer. Answer all 10 without pausing to double-check.

**Q20.** What comes next? 2, 4, 8, 16, ?

- A. 20
- B. 24
- C. 30
- D. 32

**Q21.** If all analysts are graduates and some graduates are programmers, which statement is definitely true?

- A. All analysts are graduates.
- B. All graduates are analysts.
- C. All programmers are analysts.
- D. No analyst is a programmer.

**Q22.** A report is due today. You notice one number may be incorrect, but your manager needs the report immediately. What is the best first action?

- A. Ignore it.
- B. Submit without checking.
- C. Quickly verify the critical number and flag uncertainty if needed.
- D. Delete the number.

**Q23.** Choose the odd one out.

- A. 12
- B. 18
- C. 24
- D. 31

**Q24.** If  $A=1$ ,  $B=2$ ,  $C=3$ , what is the value of  $CAB$ ?

- A. 5
- B. 6
- C. 7
- D. 8

**Q25.** Which is the strongest approach when two data sources disagree?

- A. Pick the larger number.
- B. Use the source you opened first.
- C. Check definitions, timestamps, and data quality before deciding.
- D. Average the two values automatically.

**Q26.** What is 15% of 200?

A. 20

B. 25

C. 30

D. 35

**Q27.** Complete the pattern: AB, CD, EF, ?

A. GH

B. GI

C. FG

D. HI

**Q28.** If yesterday was Monday, what day is tomorrow?

A. Monday

B. Tuesday

C. Wednesday

D. Thursday

**Q29.** When solving a difficult question in a timed test, what is generally the best strategy?

A. Spend unlimited time on it.

B. Skip immediately every time.

C. Estimate the cost of time and return later if appropriate.

D. Change answers randomly.

## **Section H — Reading Comprehension**

A data analyst rarely works with perfectly clean information. Sales records may contain missing values, duplicate transactions, inconsistent date formats, or figures that do not match another system. The analyst's job is not simply to calculate an answer. First, the analyst must understand what each field represents, check whether the data is complete and consistent, and identify problems that could affect the conclusion. A useful analysis should also be explainable to a business stakeholder. If a dashboard shows that sales increased, for example, the analyst should be able to explain which period was compared, which transactions were included, and whether any unusual data issue could have influenced the result. This combination of technical accuracy and business understanding is what makes analysis useful for decision-making.

**Q30.** What is the main idea of the passage?

A. Analysts should only learn programming.

B. Data analysis requires both data quality checks and business understanding.

C. Dashboards are more important than raw data.

D. Missing values should always be deleted.

**Q31.** According to the passage, what should an analyst do before drawing a conclusion?

A. Immediately create a dashboard.

**B.** Ignore inconsistent records.

**C.** Understand the data and check its quality and consistency.

**D.** Only calculate the average.

**Q32.** Why should an analyst be able to explain which transactions were included?

**A.** To make the dashboard colorful.

**B.** To make the analysis transparent and support the validity of the conclusion.

**C.** To avoid using SQL.

**D.** To reduce the number of columns.

### Answer Sheet — Record only your options

|  |           |           |           |           |           |
|--|-----------|-----------|-----------|-----------|-----------|
|  | Q3: ____  | Q4: ____  | Q5: ____  | Q6: ____  | Q7: ____  |
|  | Q11: ____ | Q12: ____ | Q13: ____ | Q14: ____ | Q15: ____ |
|  | Q19: ____ | Q20: ____ | Q21: ____ | Q22: ____ | Q23: ____ |
|  | Q27: ____ | Q28: ____ | Q29: ____ | Q30: ____ | Q31: ____ |

Diagnostic note: Day 1 is intended to establish your baseline. Do not worry about the score; accuracy and mistake patterns will guide later practice.